

Consider the wheel again. If you are a Stone Age historical thinker called on to predict the future in a comprehensive report for your chief tribal planner, you must project the invention of the wheel or you will miss pretty much all of the action. Now, if you can prophesy the invention of the wheel, you already know what a wheel looks like, and thus you already *know how* to build a wheel, so you are already on your way. The Black Swan needs to be predicted!

But there is a weaker form of this law of iterated knowledge. It can be phrased as follows: *to understand the future to the point of being able to predict it, you need to incorporate elements from this future itself*. If you know about the discovery you are about to make in the future, then you have almost made it. Assume that you are a special scholar in Medieval University's Forecasting Department specializing in the projection of future history (for our purposes, the remote twentieth century). You would need to hit upon the inventions of the steam machine, electricity, the atomic bomb, and the Internet, as well as the institution of the airplane onboard massage and that strange activity called the business meeting, in which well-fed, but sedentary, men voluntarily restrict their blood circulation with an expensive device called a necktie.

This incapacity is not trivial. The mere knowledge that something has been invented often leads to a series of inventions of a similar nature, even though not a single detail of this invention has been disseminated—there is no need to find the spies and hang them publicly. In mathematics, once a proof of an arcane theorem has been announced, we frequently witness the proliferation of similar proofs coming out of nowhere, with occasional accusations of leakage and plagiarism. There may be no plagiarism: the information that the solution exists is itself a big piece of the solution.

By the same logic, we are not easily able to conceive of future inventions (if we were, they would have already been invented). On the day when we are able to foresee inventions we will be living in a state where everything conceivable has been invented. Our own condition brings to mind the apocryphal story from 1899 when the head of the U.S. patent office resigned because he deemed that there was nothing left to discover—except that on that day the resignation would be justified.*

Popper was not the first to go after the limits to our knowledge. In Germany, in the late nineteenth century, Emil du Bois-Reymond claimed that *ignoramus et ignorabimus*—we are ignorant and will remain so. Somehow his ideas went into oblivion. But not before causing a reaction: the mathematician David Hilbert set to defy him by drawing a list of problems that mathematicians would need to solve over the next century.